

MAINTENANCE RUN SHEET

SCADA Simulation & System Anomaly Test Protocol

#	YOU SAY (VOICE INTERACTION)	YOU CLICK (SYSTEM ACTION)	DASHBOARD REACTS	AGENT (FROM LIVE DATA)
1	"Hey, how is everything running today?"	—	ENERGIZED, ~178 bar	<i>Calls SCADA → "Running smoothly, within normal limits..."</i>
2	"I need to do maintenance on the caster. Can you list the available SOPs?"	—	—	<i>Lists the 4 SOPs</i>
3	"Let's execute the W and S Unit Hydraulic Valve Replacement. Walk me through it."	—	—	<i>Reads Step 1, then Step 2 (LOTO)</i>
4	<i>(interrupt)</i> "Wait, stop. Before I apply lockout, what's the live pressure on the main unit?"	—	still ~178 bar	<i>Halts, reads SCADA → "~180 bar, still energized"</i>
5	"Copy. I'm executing the lockout tagout now. Watch the pressure drop."	APPLY LOCKOUT / TAGOUT	pressure ramps 178 → 0	<i>Reads SCADA → confirms dropping to zero, resumes at Step 3</i>
6	"Accumulator is isolated. Moving to step 4 — bleeding residual line pressure via the manifold dump valve. Turning it open now."	⚠ INJECT ANOMALY (click as you say "open now")	spikes past 200 bar mode = ALARM Profibus FAULT	<i>Reads SCADA → INTERRUPTS: "Stop! Pressure is spiking, close the dump valve."</i>

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7	"Valve closed! The gauge shot past 200 bar. What's going on?"	—	still high	<i>Explains likely bypass failure, tells you to step back, do NOT proceed / do NOT disconnect M12 connectors</i>
8	"I'm stepped back and clear of the manifold block."	(optional)  or just wait ~30s	anomaly auto-clears → LOCKED_OUT	<i>Confirms, says it's logging the anomaly & notifying the supervisor</i>

BEHAVIORAL NOTE

The anomaly **auto-clears after 25–35 seconds** on its own (randomized). During this window, you can either let it self-resolve (narrate *"the supervisor reset the isolation block"*) or manually hit **Reset System**.